

Reflection Statement

Throughout this semester, I played a key role in the development of the backend for our QueueMaster system, specifically focusing on creating and managing the booking functionality for vehicles. My main responsibility involved designing and implementing the BookingController using Spring Boot, which served as the core API layer that connects the frontend with the database. This experience significantly improved both my technical and problem-solving skills, as well as my understanding of collaborative software development.

One of my primary contributions was building RESTful endpoints that handle vehicle bookings. I implemented endpoints such as GET, POST, PUT, and DELETE to allow users to view bookings, create new ones, update vehicle information, and complete washes. This required a solid understanding of how HTTP methods map to CRUD operations and how data flows between the client and server. For example, I designed the addBooking method to automatically assign statuses and calculate payments based on the wash type. This reinforced my ability to translate business rules into code efficiently.

Another important aspect of my work involved integrating repositories using Spring Data JPA. By using VehicleRepository and SaleRepository, I was able to interact with the database without writing complex SQL queries. This abstraction made it easier to focus on application logic rather than database management. However, I still had to understand how entities relate to tables and how data persistence works. Through this, I gained a better understanding of ORM (Object-Relational Mapping) and how Spring Boot simplifies backend development.

I also implemented business logic related to the progression of vehicle statuses. Vehicles move from "In Queue" to "Washing" and finally to "Ready." Designing the moveToNextStatus method taught me how to manage state transitions and ensure that the system behaves predictably. I learned the importance of handling edge cases, such as invalid states, and ensuring that data remains consistent throughout operations.

An important feature I developed was the ability to complete a wash and record sales. When a vehicle is marked as complete, the system transfers relevant data into a SaleRecord entity and removes the vehicle from active bookings. This functionality helped me understand how to manage data lifecycle and maintain historical records. Additionally, I implemented the getDailyTotal endpoint, which calculates the total revenue for the current day. This required me to work with Java streams and perform aggregation operations, which strengthened my knowledge of functional programming in Java.

One challenge I encountered was ensuring that the system handled null values and unexpected inputs properly. For example, I had to safely process the washType field to avoid null pointer exceptions. This taught me the importance of defensive programming

and validating inputs before using them. I also learned how small oversights can lead to runtime errors, which pushed me to write cleaner and more robust code.

Another challenge was integrating the backend with the frontend. Since the frontend was hosted on a different port, I needed to configure CORS using `@CrossOrigin`. This was my first experience dealing with cross-origin resource sharing, and it helped me understand how browsers enforce security policies and how servers can allow controlled access. Debugging API calls and ensuring that endpoints worked correctly with the frontend taught me how to use tools like Postman and browser developer tools effectively.

Working as part of a group also enhanced my collaboration skills. We had to coordinate our work to ensure that the frontend and backend integrated smoothly. This required clear communication, version control using Git, and proper documentation of our APIs. I learned how to manage merge conflicts and how important it is to write clean, readable code so that team members can easily understand and build upon it.

From a technical perspective, I gained hands-on experience with Spring Boot, REST APIs, and backend architecture. Before this project, my understanding of backend development was mostly theoretical. This project allowed me to apply those concepts in a practical setting. I also improved my Java programming skills, particularly in areas such as exception handling, working with collections, and using annotations to simplify configuration.

In addition to technical skills, I developed problem-solving and critical thinking abilities. When something did not work as expected, I had to debug the issue by tracing the flow of data and identifying where things went wrong. This often involved checking logs, analyzing API responses, and testing different scenarios. Over time, I became more confident in identifying and fixing issues independently.

Overall, this project was a valuable learning experience that helped me grow as a developer. I now have a better understanding of how backend systems are designed and how they interact with other components of a full-stack application. I also learned the importance of writing maintainable code, handling real-world scenarios, and collaborating effectively with a team.

Moving forward, I would like to deepen my knowledge of backend development by exploring more advanced topics such as authentication, security, and scalability. I believe the skills I gained during this project provide a strong foundation for future development work.

In conclusion, my role in developing the booking controller allowed me to make a meaningful contribution to the project while gaining practical experience in software

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development. This semester has been both challenging and rewarding, and it has strengthened my interest in pursuing a career in software engineering.